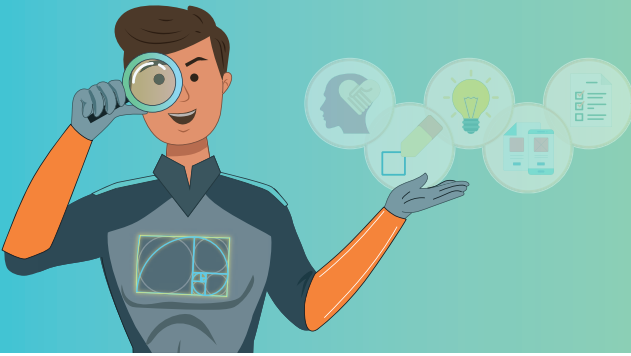


Design Thinking

04



Do you know how experts first understand people's needs before designing a product or service that enhances user experience? They use Design Thinking.



In our daily lives, we make use of many products and services. Some of them are physical, like bottles, headphones, cars etc. and some are digital, like Computer Softwares, User Interface on touch screens, control panels etc. To make use of a product or a service, we have to interact with them.

If we don't understand how to use a product or software, it is of no use to us.

When we create a product, app, website, or anything else to be used by a consumer, it is our top priority to design it in a way such that the product looks attractive, is easy to use, and is fully functional.

What is design?

Design is the process of imagining and planning the creation of products, buildings, systems, services, etc. which address a specific purpose. Design is centred around users and it is used to create solutions for people.

Discuss the design features and functions of the following objects:

- Who is the potential user?
- What materials would be used to make them? And why?
- What features do they have?
- What features should be added to them?



Wheelchair



School bag

Activity 1

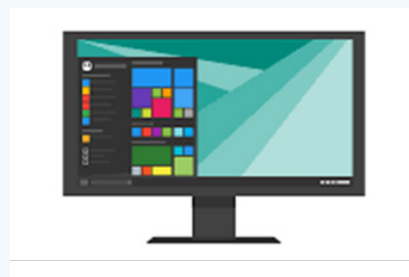
- List down any 5 devices which perform tasks and give a result after we provide an input.

Sample Answer:

1 Smartphone



2 Computer



Activity 1

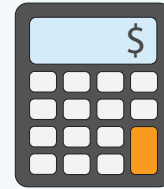
3 Smart watch



4 Smart TV

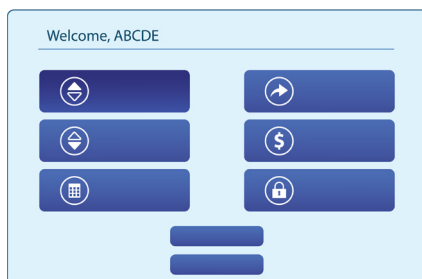


5 Calculator



All the listed devices are designed and developed for humans, who are the users. They are designed so that people can easily interact with them and understand their workings.

Users interact with machines through input and output devices. These are called the **User Interface (UI)** of the machine.



ATM machine UI



Digital switch UI



Digital dashboard of a car UI

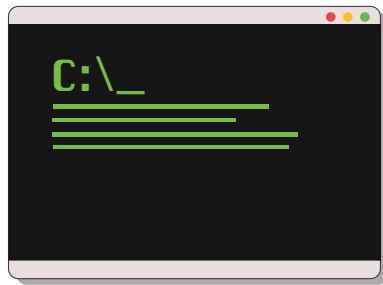
User Interface (UI)

In the digital world, a user interface (UI) is the means by which a user interacts with and controls a software application or hardware device.

The UI is basically used by an application for the following:

- To get an input or information from the user
- To send inputs from the user to the program for processing
- To display results on the screen after processing

Earlier, the UI on computers was mainly text-based, known as Command Line Interface (CLI). Today, most digital devices like smartphones, computers etc. use graphical elements in their UI, known as Graphical User Interface (GUI)



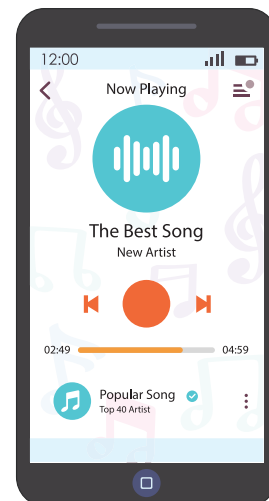
Graphical User Interface (GUI)

A **Graphical User Interface (GUI)** is a user interface that includes graphical elements such as windows, icons, and buttons in addition to text. A major advantage of GUIs is that they make computer operations easier to use and understand, while looking attractive.

For example:



Weather app UI



Music Player app UI

4. Design Thinking

Important aspects of a well-designed UI for an application or website:

- Consistency in functionality - functions, appearance and system terminology
- Error prevention and simple error handling
- Simple and pleasing design

Use Conventional Elements

Maintain Uniformity

Employ the Right UI and UX designer

Make it as interactive as possible

Consider Platform Specific Designs



Don't hamper the speed of the app

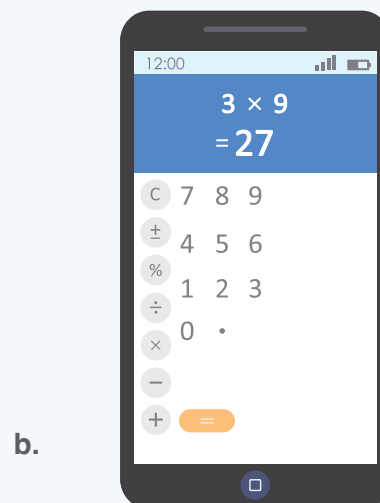
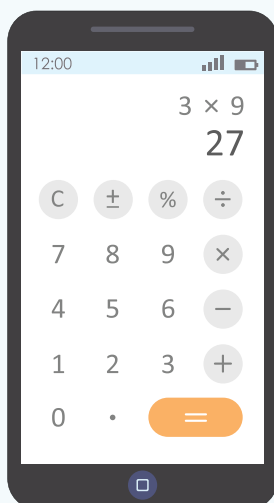
Keep it simple and don't complicate things

Keep the focus on limited things

Have detailed understanding of your audience

Activity 2

Identify which of the following GUIs is better and discuss why.



Answer:

GUI – (A) is better than GUI – (B) because the first one has clear visibility of the inputs and display, better utilisation of available space and it is easy to understand. It has all the features required for an effective UI and for a good User Experience (UX).

User Experience (UX)

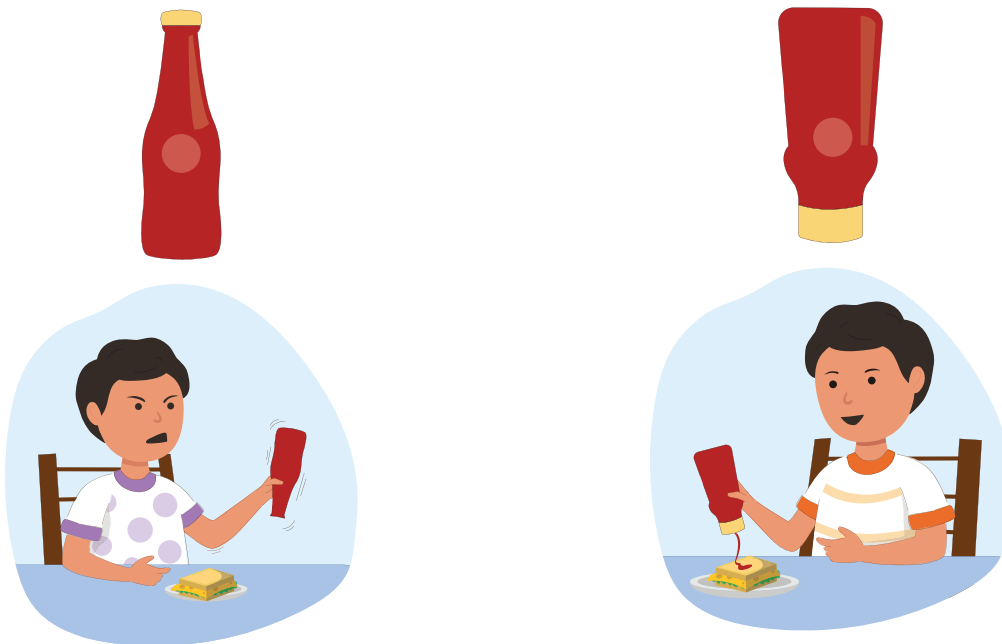
User Experience (UX) is the overall experience of a person using a product or application, especially in terms of how easy or pleasing it is to use. A good UX should:

- Be simple to use without any training required to operate
- Not take much time to display results
- Be efficient and not have unnecessary processes
- Have an attractive appearance
- Be enjoyable to use
- Be created after detailed research about the user's requirement

When these conditions are met, the user experience (UX) is better.

Examples of UI and UX:

A ketchup bottle is a good example of UI and UX.



Product designs with their respective user experiences

- These are the products that the user will interact with to add ketchup from a bottle to their food.
- **Left bottle:** Many users find it difficult to get ketchup out of the left bottle. This design still serves the purpose of storing ketchup in a container but it does not provide a good user experience.
- **Right bottle:** Users find it easy to get ketchup out the right bottle by squeezing the bottle. This design serves the purpose of storing ketchup in a container and provides a better experience for the user.

Exercise

Fill in the blanks

- 1 _____ controls the display for the user on a computer monitor or a smartphone screen.
- 2 A _____ is the overall experience of a person using a product or application.

In this lesson, we will use the concepts of UI and UX to design an app of our own while following the design process.

The design process is the same for any physical product or any virtual product (like applications, games, webpages) on digital platforms like mobiles, laptops or the web.

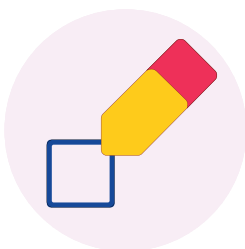
Note:

For simplification, we will discuss the mobile app design process. However, the same process can be applied for any product design with minor modifications.

The basic design process consists of the following steps:



Empathise



Define



Ideate



Prototype

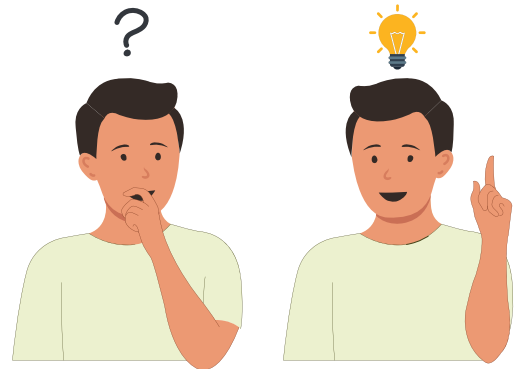


Test

1. Empathise

In this phase, we learn about the user who is going to use our product or service.

- Who is the user?
- Which age group do they belong to?
- What do they do?
- What are they going to use the product for?
- What are their shortcomings?
- What problems do they face with similar products?
What functionality are they most interested in?



For example:

To be able to perform difficult mathematical calculations easily, we can create a calculator app.

A survey should be an efficient method to gather as much information as possible from potential users. There are many online survey tools as well as offline methods of interviewing and surveying on paper through which this information can be gathered.

2. Define

“What do we need to create to fulfil the user’s needs?”

- This is the question that needs to be answered in this phase of the design process. We can answer it once we have narrowed down who the users are and what application do they need.
- As the application concept gets finalised, identify what a user might need from it.

For example:

- The app should be simple to understand and use.
- It should be accessible in offline or online mode.
- The screen should be compatible with all devices.
- The app should match the expected functionality.
- The app should not ask for any unnecessary personal information from the user.



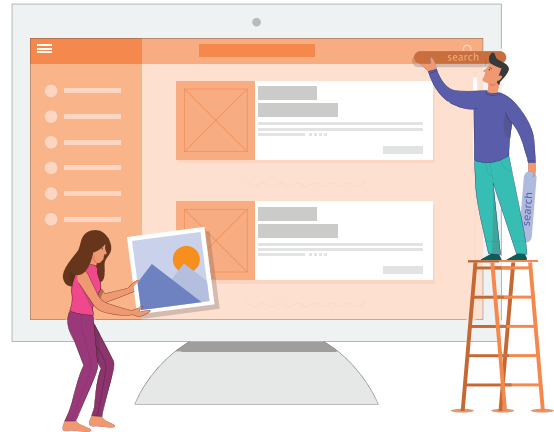
3. Ideate

In this process, we draw rough drafts of the product and brainstorm which design will lead to the best results.

We also have to consider the cost of production, supply, ease of using the product and aesthetics (how it looks) while designing the drafts.

User Interface:

- A simple design will keep the user at ease while interacting with the product.
- There should not be too many elements placed randomly on a product or application.
- Forms to retrieve user information should be as short as possible.
- Space, colour, shapes, text fonts should be well designed, easily readable and comforting to the eyes

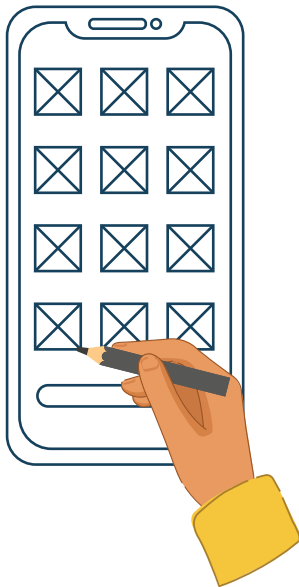


We learnt that the process of design thinking focuses on user.
Let's use this process to solve user problems.



4. Prototype

- In this activity, we will draw prototypes of a product. We will also brainstorm which design will provide the best user experience.
- The best practice is to design a mock layout of the product on paper to get a clear idea of what the product should look like.



5. Implement functionality for the app

- After completing the app design process discussed above, the next step is to implement the functionality of the product (in this case an app) as per user requirements.

6. Test

- In this phase, we take a prototype to the users. We ask them to use it and give feedback.
- Through the feedback, we can assess which features are expected, understand our drawbacks and redesign the product to improve it further.

Activity 3

Using the design thinking process, design a layout on paper for an app to remind the user to drink water as per a set schedule.

1 Ideate an app concept to provide a solution for the user to drink water at regular intervals

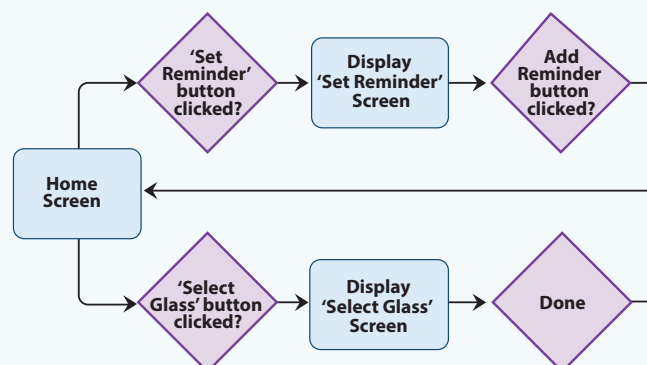
- Drinking water at regular intervals helps us stay hydrated. It can also help to prevent common problems such as stomach pain, fatigue, overeating etc.
- But due to our busy schedule we tend to forget to drink water regularly.
- To avoid that, we can design an app that will keep reminding us to drink water as scheduled.

2 Understand user requirements

- Here, the user requirement is to drink water at regular intervals from 7 a.m. to 9 p.m.
- The app should be simple to understand and use.
- The user should be able to select different glass sizes.
- It should display a daily log of water consumption.
- The screen should be compatible with all mobile phones.
- The app should perform the expected functionalities.

3 Plan the flow of the app

- There should be a feature to set the time and ring the alarm on time.
- Once the alarm rings, show a message or an image as a reminder to drink water.
- 'Switch off the alarm' and 'Select size of the glass' buttons should be available.
- After selecting the size of the glass, the user should drink a glass of water.
- The user will press the 'Done' button once they drink water.
- Add the quantity of water to the daily log and display it on the home screen.

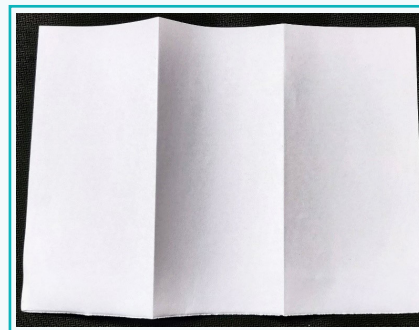


Activity 3

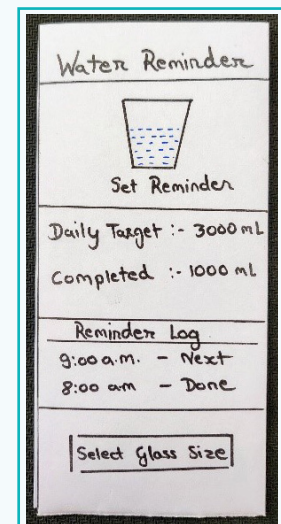
4 Design a layout on paper

The images given below are for reference. Try to create unique designs of your own.

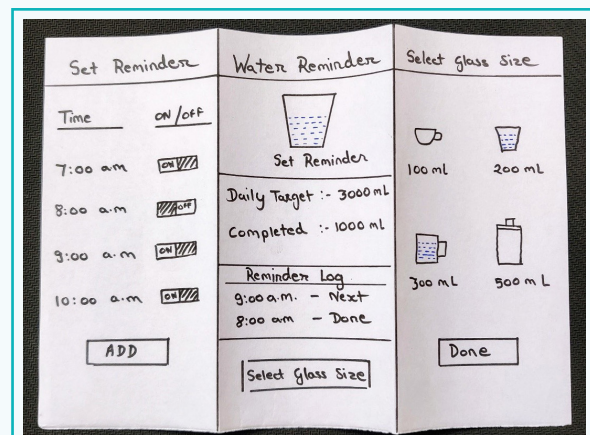
a. Fold the half-size A4 paper thrice.



b. Design and draw the user interface for the first screen.

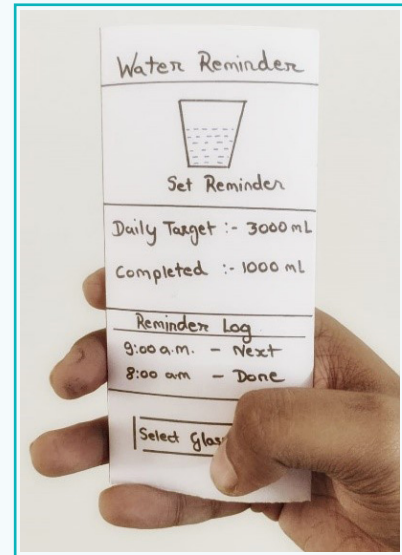


c. Design and draw the user interface for the second and third screen.



Activity 3

d. Try to handle the designed screen with one hand.



Our design is now ready! Now, the next steps would be to implement the functionality and test it.

Exercise

State whether True or False:

- 3** Feedback on the prototype of product or service from the users helps to improve the product.
▲ _____
- 4** Design is the process of making a product look good and plan its functionality while maintaining a good user experience.
▲ _____

In this lesson, we learnt about the importance of design thinking. We also learnt about the various steps involved in the design thinking process. The steps involved are empathise, define, ideate, prototyping and testing. Through this process, it will be interesting to create applications that can truly create value and benefit everyone.

Users and their problems are the main focus areas for design thinking. Learning more about users helps us create better solutions for them.



Tech Flash

- **Design** : Design is the process of imagining and planning the creation of products, buildings, systems, services, etc. which address a specific purpose.
- **User Interface (UI)** : A user interface (UI) is the means by which a user interacts with and controls a software application or hardware device.
- **User Experience (UX)** : UX is the overall experience of a person using a product or application, especially in terms of how easy or attractive it is to use.